

TECHNICAL DOCUMENTATION

# Sustainability Dashboard

Data Flows & Business Formulas

Mill Onboarding · Supplier Due Diligence · Traceability  
End-to-end mapping across modules and Google Sheets

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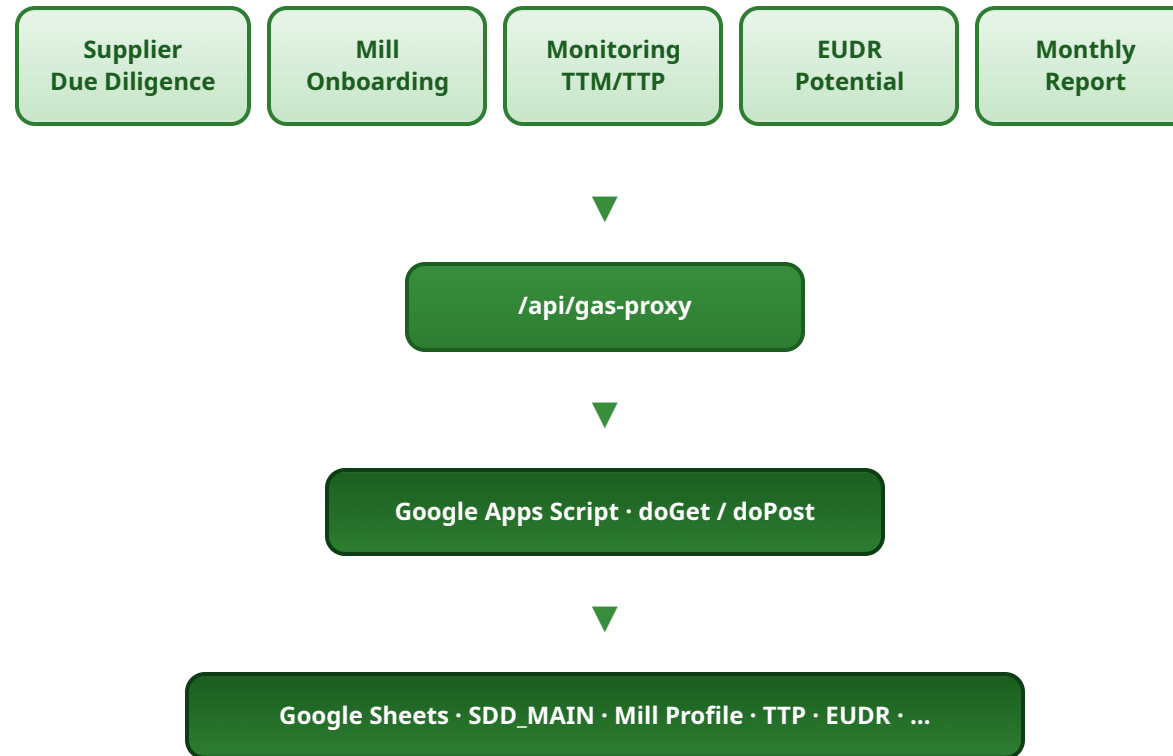
19 slides — complete data flow reference

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# 01 · System Architecture

Frontend → API Proxy → Google Apps Script → Google Sheets



**Key principle:** Mill scores and risk levels are computed in Google Sheets. The web app never overwrites formula columns.

# 02 · Supplier Due Diligence (SDD)

Panel #panel-supplier-dd

- **STORAGE** SDD\_MAIN · SDD\_MILL\_LIST · SDD\_FFB\_LIST
  - **KEY** submission\_id (SUB-YYYYMMDD-XXXXXX)
  - **TYPE** MILL · KCP · TRADER
  - Excel upload — Main Form + Traceability A/B
  - Screening form — SCR, GRV, PRI fields
- Approve / Hold / Reject (approver decision)
  - NBL check vs NBL & Unilever NBL sheets
  - Export PDF screening report
  - TML / FFB spatial screening (modals)
  - Legacy: flat SDD Data sheet (still supported)

## 03 · SDD Lifecycle

State machine — from import to onboarding complete



# 04 · SDD Data Destinations

| Trigger                  | Target Module           | Backend Function                     |
|--------------------------|-------------------------|--------------------------------------|
| statusSDD = APPROVED     | Contact List Supplier   | syncContactFromApprovedSubmission_() |
| APPROVED + Submitted     | Mill Task List          | isApprovedSubmittedSddRow_()         |
| Save mill from Task List | Mill Onboarding Profile | add / update sheet mill              |
| mill_ttp_sync payload    | Monitoring TTP/TTM      | syncTtpFromMillOnboarding_*          |
| NBL check                | SCR - No Buy List       | runSddNblCheck()                     |
| Report period            | Monthly Report \$01     | fetchSddList()                       |

# 05 · SDD → Mill Onboarding Mapping

mapSddRowToMillPayload() / mapSddTmlRowToMillPayload()

| Mill Column              | SDD Source                             |
|--------------------------|----------------------------------------|
| GROUP NAME               | Group Name                             |
| COMPANY NAME             | Company Name                           |
| MILL NAME                | Mill Name / KCP Name / TML - Mill Name |
| COORDINATES              | Latitude + Longitude (comma-separated) |
| MILL CAPACITY (TON/HOUR) | Mill Capacity (Ton/Hour)               |
| SOURCE TYPE              | MILL / KCP / TRADER                    |
| TRADER NAME / UML ID     | Trader parent / TML - UML ID           |

# 06 · FFB → TTP/TTM Mapping

One TTP row = one FFB supplier + Main Form context

| TTP Column                    | SDD / Mill Source              |
|-------------------------------|--------------------------------|
| GROUP / COMPANY / MILL NAME   | Mill Onboarding identity       |
| FFB SUPPLIER GROUP / NAME     | FFB - Supplier Group/Name      |
| CATEGORY · LAT/LONG · VILLAGE | FFB location & category fields |
| CONCESSION / PLANTED AREA     | FFB area fields                |
| LEGALITY · ISPO/RSPO/ISCC     | FFB certification              |
| FFB SUPPLY to MILL (TON)      | FFB - Total Supply FFB (Ton)   |

Full reference: [docs/sdd-to-ttm-ttp-field-mapping.md](#)



# 07 · Mill Onboarding — Entry Points

Mill Profile (CPO/PK) · Mill Waste (POME/SHELL)

- **Manual Add/Edit**

Modal #btn-add-mill

- **SDD Task List**

"+ Add to Mill" from approved screening

- **Supply Excel Import**

Upload → draft → submit

- CPO/PK → **mill** sheet

- POME/SHELL → **millWaste** sheet

- KCP plant → **FACILITY NAME PK**

- Refinery → **FACILITY NAME CPO**

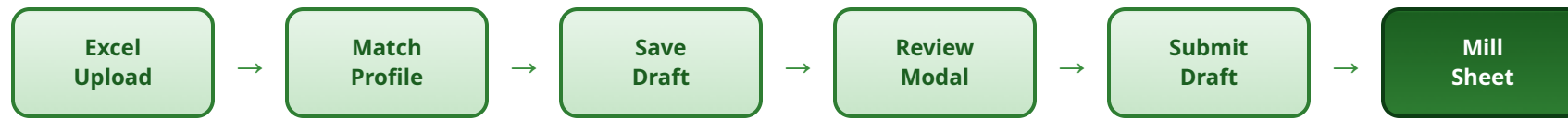
- Staging: **Supply Import Draft**

# 08 · Mill Onboarding → Outputs

| Consumer                 | Join Key                     | Data Used                  |
|--------------------------|------------------------------|----------------------------|
| Monitoring TTP/TTM       | GROUP   COMPANY   MILL   UML | Identity + FFB sync        |
| Monthly Report §02       | Period + risk level          | High-risk MILL rows        |
| Facility Performance     | FACILITY NAME CPO/PK         | Supply qty, traceability % |
| Questionnaire Monitoring | group   company   mill       | Mill registry              |
| EUDR Potential           | group   company   mill       | Identity sync              |
| Mill Executive Report    | Quarter                      | Supply, NBL, certification |

# 09 · Supply Task List

Embedded in the Mill Onboarding panel



- match\_status: **matched** · **new** · group\_mismatch
- Formula columns restored from row above — never overwritten
- Repair drift: **reconcileSupplyDraft\_()**

# 10 · Google Sheet Formulas (Read-Only)

MILL\_FORMULA\_HEADERS\_ — never written from the web app

● SCORE · TOTAL SCORE · LEGALITY SCORE

● DEFORESTATION / BURN / PEAT SCORE

● TOTAL GRIEVANCES · POLICY · CERTIFICATION

● RISK LEVEL · RESULT RISK LEVEL

● BUYER NO BUY LIST · SUPPLIER STATUS

● PERCENTAGE SUPPLY CPO/PK/ISCC/INS/SHELL

● PRODUCT SUPPLY · STATUS SUPPLY CPO/PK

● TOTAL POME SUPPLY · MAX/REMAINING STOCK

● TOTAL SCORE SUPPLY · TOTAL RISK LEVEL

● SD Monitoring: DAY LEFT · Status · Result

# 11 · App Formulas (Frontend)

Computed in the web app — not in Google Sheets

## Grievance Risk Score

Total = sum of 6 indicators (score 1–3)  
≤ 8 → Low · ≤ 11 → Medium · > 11 → High  
*src/grievance-risk.js*

## TTP Aggregate %

$\Sigma(\text{traceable vol}) / \Sigma(\text{supply}) \times 100$   
Fallback: avg of % TRACEABLE columns  
*ttpAggregateTotalTraceablePct\_()*

## TTM Coordinate %

Pool: MILL/TRADER/REFINERY + supply > 0  
 $\% = \Sigma(\text{traceable qty}) / \Sigma(\text{total qty}) \times 100$   
*ttpCalcTtmCoordinatePct\_()*

## EUDR Potential Status

All enabled criteria pass → Potential  
Any fail → Not Potential  
*eudrComputeStatus\_()*

# 12 · Grievance & Mill Risk Reason

● **Grievance** — all 6 indicators required before classification

- Publish · Subject · Repeat · Consequence
- Group Scale · No Response

● **Mill Risk Reason** — UI pills derived from field gaps

- No Coordinate · Legality Not Complete
- Non APL · No NDPE · No Certification
- Deforestation > 0 ha · On No Buy List

# 13 · Module Connection Map



# 14 · Google Sheets Registry

| Key              | Tab Name                        | Module          |
|------------------|---------------------------------|-----------------|
| mill / millWaste | Mill Onboarding Profile / Waste | Mill Onboarding |
| sddMain/Mill/Ffb | SDD_MAIN / MILL_LIST / FFB_LIST | Supplier DD     |
| ttp              | Monitoring TTP/TTM              | Traceability    |
| grievance        | Grievance Monitoring            | Grievance       |
| eudrPotential    | EUDR Potential                  | EUDR            |
| contactSupplier  | Contact List Supplier           | Contact List    |



# 15 · SDD vs EUDR DDS

Two separate modules — no direct coupling

| Aspect  | SDD (Supplier Due Diligence)    | EUDR DDS                       |
|---------|---------------------------------|--------------------------------|
| Panel   | #panel-supplier-dd              | #panel-due-diligence-statement |
| Storage | SDD_MAIN + child sheets         | EUDR DDS + child sheets        |
| Purpose | Internal screening + onboarding | EU EUDR export package         |
| Output  | Task List → Mill → TTP          | DOCX/PDF for buyers            |

# 16 · Validation Rules

Guard rules that prevent data inconsistency

- Submitted SDD cannot be downgraded to Draft
- Task List: SCR = submitted AND statusSDD = APPROVED
- TTP sync: full identity (GROUP/COMPANY/MILL/UML)
- Supply submit: match\_status = matched or new
- Mill save: formula columns stripped before write
- EUDR: all enabled criteria must pass
- Grievance: all 6 indicators required before classification
- Coordinates: normalizeCoordinate() + recoverCoord()

# 17 · Code Reference

| Area                | File                                             |
|---------------------|--------------------------------------------------|
| Backend CRUD + sync | scripts/GoogleAppsScript-backend-v3-full.gs      |
| Main app logic      | src/main.js                                      |
| Grievance risk      | src/grievance-risk.js                            |
| Mill risk reason    | src/mill-risk-reason.js                          |
| Monthly report      | src/monthly-report-ui.js · monthly-report-pdf.js |
| TTP field mapping   | docs/sdd-to-ttm-ttp-field-mapping.md             |

# Thank You

Sustainability Dashboard — Data Flow Documentation